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Quantifying Acid System Effectiveness Using Microseismic and 3D Sonic Imaging in Kuwait

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Abstract

Stimulation performance in tight carbonate formations remains highly sensitive to fluid selection and placement. This work investigates the effectiveness of acid systems—28% Hydrochloric Acid (HCl) and a 28% Lightweight Single-phase Retarded Acid System (SPRA)—in horizontal wells drilled in a complex tight carbonate reservoir of Kuwait's Wafra Field. Key objectives include characterizing fracture geometry, evaluating vertical growth into overlying barriers, and analyzing production behavior as a function of lateral landing strategy and fluid design using advanced diagnostics.

A 23-stage Plug and Perforation acid fracturing program was executed using mechanical diversion and real-time data acquisition to improve stimulation efficiency. Microseismic monitoring and far field 3D sonic imaging using azimuthal sensors provided detailed insights into fracture orientation, height, and extent. Chemical tracers were deployed to quantify contribution from each interval. Comparative analysis between HCl and Lightweight SPRA focused on fluid placement efficiency and stimulated volume. Post-treatment evaluation incorporated geomechanical modeling and seismic interpretation to validate design assumptions and guide future stimulation strategies in the target tight carbonate's structurally heterogeneous, low-permeability environment.

Fracture orientation across all observed stages aligned consistently at N45°E. Approximately 70% of the fracture propagation remained within the target zone, while 30% extended vertically into the barrier formation. Although both acid systems produced similar upward growth profiles, Lightweight SPRA-treated stages demonstrated significantly greater downward propagation and longer fracture lengths—nearly double those achieved with HCl—indicating superior penetration and reservoir contact.

Tracer analysis confirmed broader stimulation coverage in all Lightweight SPRA stages, supporting improved fluid efficiency and treatment distribution. Microseismic and 3D sonic imaging results correlated strongly with pre-job geomechanical predictions, validating model accuracy and enhancing confidence in the design workflow.

Operationally, the Lightweight SPRA system delivered more consistent performance in terms of fracture containment and connectivity in tight, variable rock properties. The findings highlight how optimized fluid

selection, coupled with integrated diagnostics, can significantly improve stimulation outcomes in carbonate systems with complex geologic features.

This case underscores the value of tailoring stimulation approaches based on real-time data and subsurface conditions, offering practical insights into acid system behavior and landing zone impacts—both of which are critical for future field development planning.

This paper documents the first application of integrated 3D sonic imaging, microseismic monitoring, and tracer analysis in an acid fracturing campaign. It provides a unique and quantitative comparison of two acid systems in a carbonate reservoir, offering new insights into how fluid design influences fracture geometry and effectiveness. The results enhance geomechanical modeling accuracy and present a data-driven workflow that can be replicated in similar tight carbonate environments to improve stimulation design and reservoir connectivity.

Introduction

Estimates show that around 50 to 60% of the world's conventional oil and gas reserves are held in carbonate reservoirs (Burchette 2012). The Middle East, for example, is dominated by carbonate fields, with around 70% of oil and 90% of gas reserves of the region held within these fields (Tavakoli et al. 2022).

Tight carbonate reservoirs are characterized by low porosity, low permeability, and complex geologic features, making them challenging to develop economically. These reservoirs require advanced stimulation techniques to enhance reservoir connectivity and unlock their full production potential. Acid fracturing is a widely used method to improve production in such formations by creating conductive pathways and increasing the stimulated reservoir volume.

This study evaluates the stimulation effectiveness of two acid systems—28% Hydrochloric Acid (HCl) and a 28% Lightweight Single-Phase Retarded Acid System (SPRA)—in a tight carbonate reservoir located in Southwest of Kuwait. The work was conducted in two horizontal wells, Well-A and Well-B, treated with a total 23-stage Plug and Perforation (PnP) acid fracturing program. Advanced diagnostic tools, including microseismic monitoring, tracer analysis, and far-field 3D sonic imaging, were used to assess fracture geometry, fluid placement efficiency, and production behavior.

Moreover, a separate study of the same tight carbonate reservoir was conducted on the vertical Well-C to update the geomechanical model using rock laboratory techniques on core plugs. These evaluations were key to identifying the landing target of the horizontal wells in the field across the thick laminated carbonate reservoir.

The novelty of this work lies in its integration of multiple diagnostic tools to quantify acid system performance and its impact on fracture geometry and reservoir connectivity. By comparing the two acid systems, this study provides valuable insights into fluid design optimization and its role in improving stimulation outcomes. The findings will inform future field development strategies in tight carbonate reservoirs, where effective stimulation is critical for economic viability.

Field Background

The studied reservoir is a tight mixed carbonate formation with significant structural heterogeneity and variable rock properties. The reservoir exhibits low permeability and porosity, with vertical barriers such as shale streaks and tight carbonate layers complicating fracture propagation and containment.

Historically, conventional acid fracturing treatments in similar formations have struggled to achieve consistent results due to challenges in fluid placement, fracture containment, and reservoir connectivity. The lack of detailed diagnostic data has limited the ability to optimize fluid selection and fracturing strategies. Moreover, the formation underneath the lower barrier is 90% water bearing and the risk of fracturing into it is significantly high based on existing data. A water breakthrough from the lower barrier would terminate the development of this tight carbonate reservoir.

This study incorporates advanced geomechanical analysis and diagnostic tools to improve the understanding of fracture behavior and enhance the effectiveness of stimulation treatments, based on which the entire field development strategy is built.

Objectives

The primary objectives of this study are:

- To evaluate the effectiveness of two acid systems, 28% HCl and 28% Lightweight SPRA, in stimulating a tight, thick, complex carbonate reservoir.
- To characterize fracture geometry and orientation using microseismic monitoring and far-field 3D sonic imaging.
- To quantify fluid placement efficiency and stimulated reservoir volume using tracer analysis.
- To assess production behavior as a function of lateral landing strategy and fluid design.
- To validate geomechanical modeling predictions and refine future stimulation strategies.

Methodology

Acoustic Structural Feature Imaging for Stimulation Staging Design

The wells under the study (Well-A and Well-B) were 6-1/8" horizontal laterals placed in potential pays in the tight carbonate formation. A slim azimuthal sonic tool with extended recording time was logged with slim wireline tractor conveyance in the laterals once they were cased with a 4.5" liner. The well was geo-steered with the help of near bit Gamma Ray image. The purpose of acquiring the 3D sonic imaging data was for the following applications:

- Extract representative compressional and shear velocities from the wideband frequency data thereby providing velocity data for porosity and rock mechanical property variation along the lateral section. The compressional velocity data was used to delineate porous and tight limestone intervals. Dynamic rock mechanical properties provided key input for geomechanical modelling.
- In the absence of detailed borehole images, extended listening time waveform data from the slim sonic tool with the azimuthal receivers was processed to characterize the azimuth, dip and extent of acoustic reflector surfaces for structural geology applications as per workflows specified by [Donald et. al, 2020](#).

Once the reflector surfaces geometry was characterized, they were classified according to their dip magnitude and orientation with the regional maximum horizontal stress (SH_{Max}) direction to extract structural feature information. Structural features with dip magnitude less than 30 deg were classified as bedding or features related to structural dip. High angle structural features with dip magnitude above 30 deg were assumed to be due to fractures or solution enhanced features typically present in this tight carbonate reservoir. The feature orientations are in good agreement from offset well borehole image data. Classification of the high angle features was done by integrating the Mechanical Earth Model (MEM) with the 3D sonic imaging data and performing Feature Compliance Analysis (FCA). FCA characterizes the stress state acting on the structural features / discontinuities based on their geometry and orientation with the in-situ stress tensor. FCA helps in determining the mechanical stability of structural features like fractures / faults during dynamic changes in pore pressure ([Rogers, 2002](#)). The observations from FCA ([Fig. 1](#)) are summarized below:

- NW-SE Strike – These structural features are expected to be non-compliant with regional SH_{Max} direction and mechanically stable.

- NE-SW Strike – These structural features are expected to be compliant with regional SH_{Max} direction and mechanically unstable / hydraulically conductive.
- NNE-SSW / N-S / E-W Strike – These structural features are expected to be partially compliant with regional SH_{Max} direction. Their mechanical stability and hydraulic conductivity are expected to be dynamic depending on the in-situ stress regime and dynamic pore pressure changes.

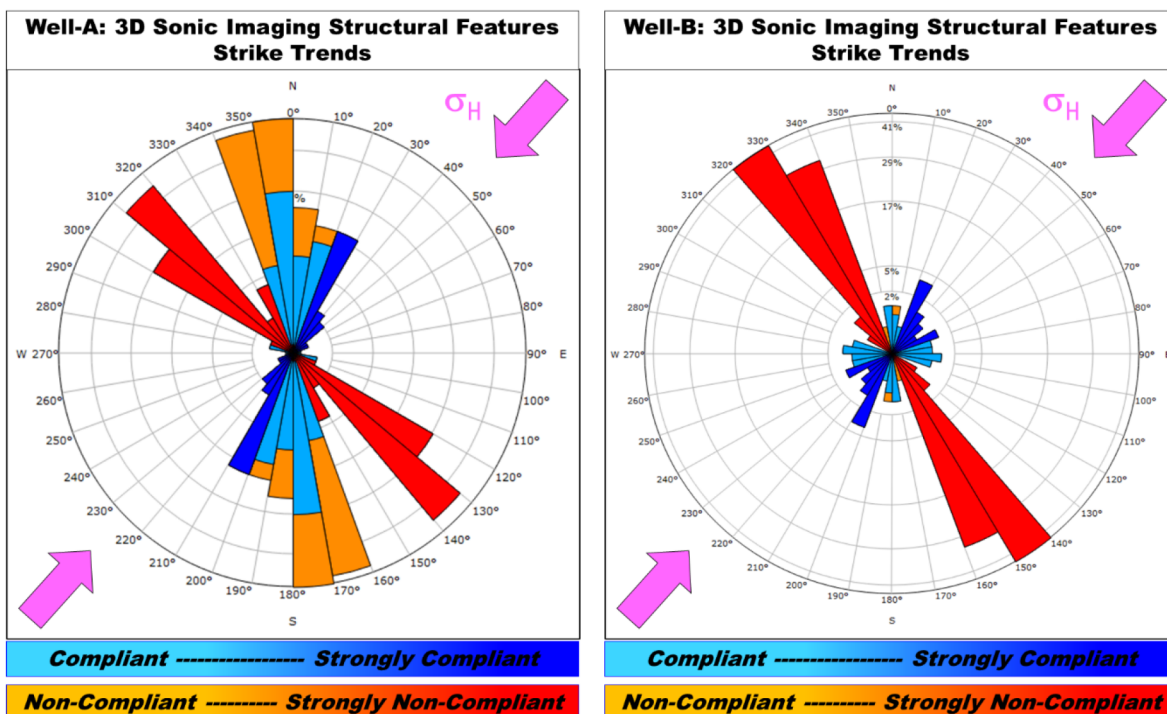


Figure 1—3D Sonic Imaging Structural Feature Strike and Compliance in Well-A & Well-B

Compliant features are expected to be open and have often been seen to be conducive for hydrocarbon production; however, they can cause unwanted fluid leak-off during well stimulation. Suitable acidization of non-compliant features can increase the reservoir stimulated volume, however their stress state can lead to high break-down pressures.

The staging design for the 2 wells was done by segmenting the lateral length into intervals with similar porosity, structural feature compliance, orientation and mechanical properties. Further segmentation was done to ensure stimulation fluid volumes were uniformly placed along the segment length after extensive modelling and simulation exercises. Perforation clusters were placed in the lowest stress intervals of each segment to ensure lower breakdown pressures to initiate acid fracturing. The final staging and perforation clusters for the wells can be seen in Fig. 2 & Fig. 3.

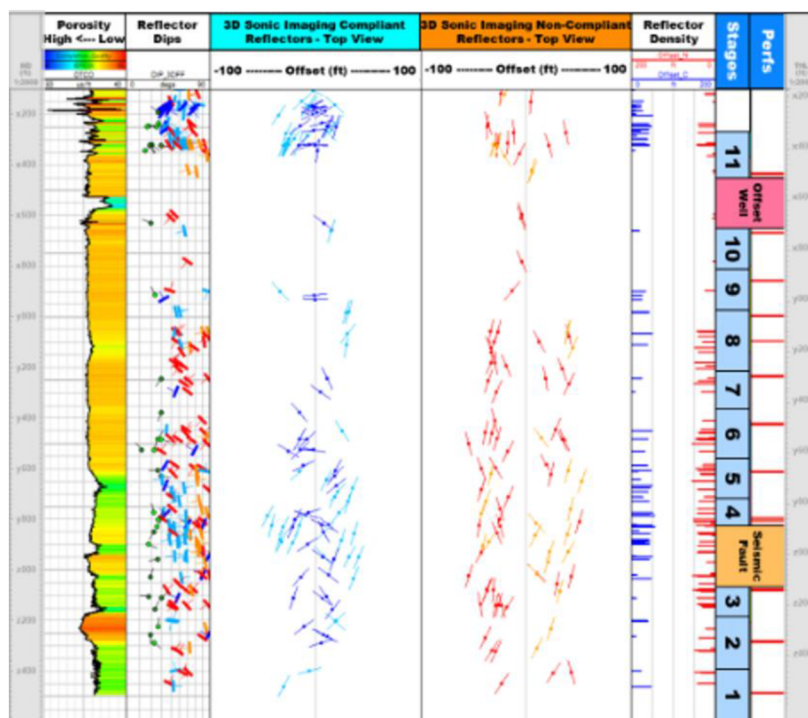


Figure 2—Staging Design in Well-A

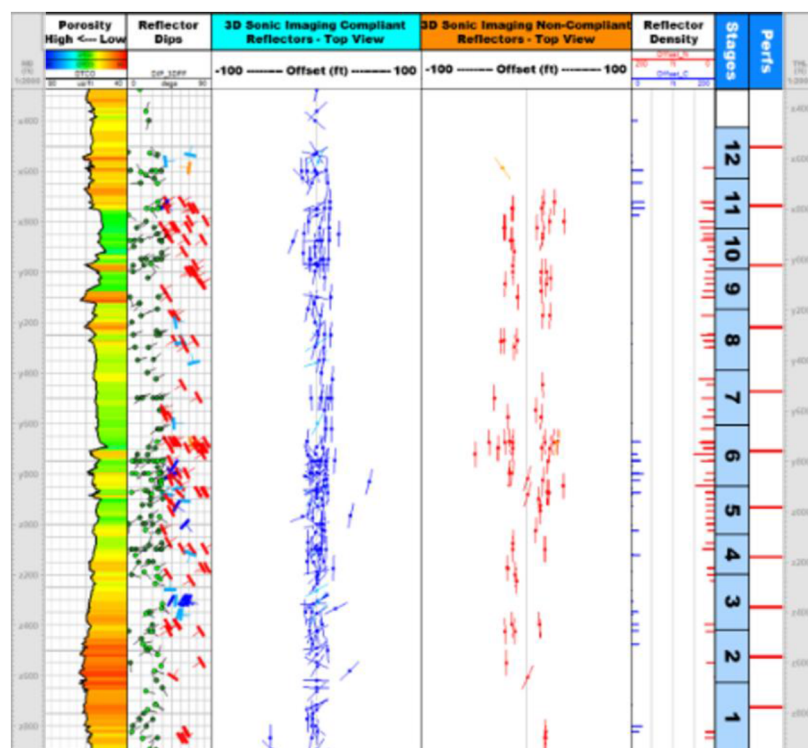
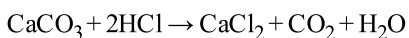


Figure 3—Staging Design in Well-B

Fracture treatment

It has been known for decades that HCl can stimulate carbonate formations to produce at higher rates. The HCl reacts with calcium carbonate to form calcium chloride, carbon dioxide, and water. The chemical equation is:



Optimum stimulation is achieved by forming wormholes that extend from the wellbore formation face into the reservoir (Shuchart, et al. 2010). Wormholes are formed when acid is pumped at a rate higher than the general rate of reaction of the acid with the formation rock. The wormholes increase the effective near-well permeability, which is important because this is where fluid velocities are highest. The net result is an increase in the productivity index (inflow rate per foot divided by pressure drawdown). HCl also helps remove filter cake from the mud used to drill the well, especially when the filter cake contains HCl-soluble materials such as calcium carbonate particles (Jackson, et al. 2011).

However, in many cases, the reaction rate of HCl is too high. This leads to limited travel distance from the wellbore and extreme facial dissolution which limits wormhole growth along the fracture face. Therefore, these fractures tend to have high near-wellbore conductivity and short etched half-lengths. To address this issue, retarded acids are often utilized. The commonly used retarded acid system is emulsified acid. Emulsified acids combine the high dissolution capacity of HCl with a slower reaction rate (Navarrete et al. 1998). This is accomplished by encapsulating micrometric acid droplets in a continuous hydrocarbon phase to form an emulsion. The major drawbacks of emulsified acids are their high viscosity, which leads to poor injectivity in tight carbonate reservoirs, high friction pressure, their complex preparation, the use of flammable diesel and high tendency to form emulsions with reservoir fluids if not tested prior to treatments.

Light Weight Single-Phase Retarded Acid System (SPRA). The development of HCl-based single-phase retarded acids has helped overcome the drawbacks of emulsified acids and improve the efficiency of acid stimulation treatments of carbonate reservoirs by allowing deeper acid penetration as observed in Fig. 4. The latest Lightweight SPRA can be formulated to reach the carbonate dissolution capacity of 28 wt% HCl, which translates to a lower volume of fluid required to achieve the desired depth of acid penetration. It has a very low viscosity, similar to the viscosity of unmodified HCl solutions and is compatible with acid friction reducers. In addition, the density of the new Lightweight SPRA is equivalent to HCl of the same acid strength which is an advancement over the previous version of Single-Phase Retarded Acid (Danican et al. 2024). This allows a relatively enhanced flowback from under-pressurized reservoirs post stimulation.

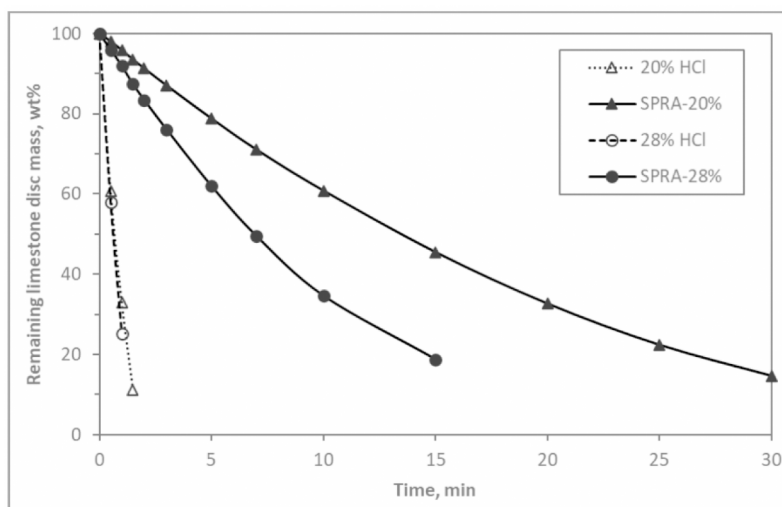


Figure 4—Limestone disc dissolution tests at 140°F (Danican et al. 2024).

Diversion Techniques. One of the critical challenges in carbonate acidizing treatments is the diversion of the acid into the lower permeability zones of the reservoir. Acid will naturally flow into the higher-permeability zones with the least resistance. Multiple chemical diversion techniques have been developed, including particulate diverter materials for near-wellbore diversion and acid-based viscous surfactant-based

diverter materials. These chemicals also can completely break and restore near-native permeability. The low-temperature particulate degradable diverter system (PDD) is developed to divert across clusters of perforations and create multiple hydraulic fractures within the same treatment stage, after which the diverter completely degrades at bottom-hole temperature.

Microseismic Monitoring

The downhole network modelling and survey design is the first step in a microseismic monitoring project. The feasibility study is vital for the success of any microseismic project as it evaluates the subsurface geophysical conditions and assesses them for microseismic detectability ranges and quality. It accounts for the effect of transmission losses, attenuation, radiation pattern of the microseismic event, detection threshold (Signal to Noise Ratio), spatial uncertainty of the events, and noise level of the surroundings (Le Calvez et al. 2015). The output of the feasibility study is the optimal sensor placement in the monitoring well which will be able to detect the largest number of high-quality events during stimulation.

For the two treatment wells in the project, the available nearby wells were evaluated for their detection range and location uncertainty. One dimensional velocity model is built from sonic logs available in the area using compressional slowness, shear slowness, and density. Using the constructed velocity model and approximate estimate of the fracture extent (laterally and vertically) with respect to the stimulation well, array position was evaluated for each potential monitoring well for detectability. Optimal array positioning is determined by identifying the configuration that exhibits the lowest moment magnitude detection. Then, the optimal array configuration is assessed for location uncertainty to understand the expected quality of the events.

Well-A had two potential monitoring wells and Well-MWA was chosen for its lower minimum moment magnitude detectability for the stimulated stages. Well-B had several potential monitoring wells; however, during the feasibility study, it was noticed that a single monitoring well could not cover the entire interval of the lateral for monitoring. Hence, two offset wells (Well-MWB1 and Well-MWB2) towards the toe of the well were chosen as monitoring wells for dual monitoring during stimulation and the first six stages were monitored for Well-B.

The microseismic configuration in both cases logged in Well-MWA, Well-MWB1 and Well-MWB2, consisted of 12 shuttles with 100 ft spacing conveyed on conventional wireline. All monitoring wells involved in the operations had been shut-in and the logging was performed riglessly. For the purpose of microseismic operations, it is imperative that the shuttles are deployed in a vertical section of the monitoring well. Furthermore, in order to achieve the most accurate microseismic detection with the least amount of background noise, necessary operational precautions must be taken by the field crews. Upon anchoring the shuttles at the required depths, the wireline cable must be clamped at the wellhead, the upper sheave wheel rigged down and the cable slacked onto the ground with the vertical section of the wireline cable being secured to the XMAS tree and wireline pressure control equipment as seen in Fig. 5. In doing so, the vibrations from the cable are drastically minimized thus almost completely removing the possibility of any background noise being recorded from the surface. Next, time-stamp and geo-coordinates need to be set up to properly and accurately map all seismic events detected by the downhole shuttles. While in Well-MWA, GPS synchronization in itself was sufficient, in Well-MWB1 and Well-MWB2, the microseismic study consisted of dual-well monitoring which required additionally to have time synchronization between the two wells and Well-B. Lastly, close coordination between the microseismic, PnP and fracturing teams is essential to ensure the shuttles are setup and have captured ample background seismic activity prior to and continuing after activating any operations-induced seismic activity, i.e. plug setting, perforation and/or acid fracturing.



Figure 5—Microseismic rig up configuration – shuttles anchored at required depth with cable slacked to eliminate surface noise

Velocity model calibration is a crucial step in microseismic acquisition and processing. The initial 1-D velocity model from logs was calibrated using the perforation shots in the treatment wells. As the location of the perforation clusters are known along with the position of array, the velocity model can be calibrated for accuracy. This calibration was performed for each stage's perforation run to ensure the robustness of the velocity model.

The microseismic data is acquired and processed in real-time during stimulation for visualization and preliminary analysis. During real-time, based on the number of events, comprehensive quality control and validation of the detected events is not always possible. For this reason, data is processed again after acquisition to ensure that the quality of the microseismic data and confidence in their event location is not compromised. The post-processed microseismic data was used for interpretation in estimation of fracture dimensions and trends observed during stimulation.

All 11 stages for Well-A were monitored with the array from Well-MWA. The initial stages were farther away from the monitoring array than the stages towards the heel and as a result, they were associated with lower event count. The first 6 stages out of 12 were monitored for Well-B with arrays deployed in Well-MWB1 and Well-MWB2. There was no considerable difference in the monitoring distances from both wells for Stage 1 and Stage 6. [Fig. 6](#) and [Fig. 7](#) indicate the monitoring setup with the distance from the center of the array to the perforations for the first and last stages monitored for Well-A and Well-B respectively.

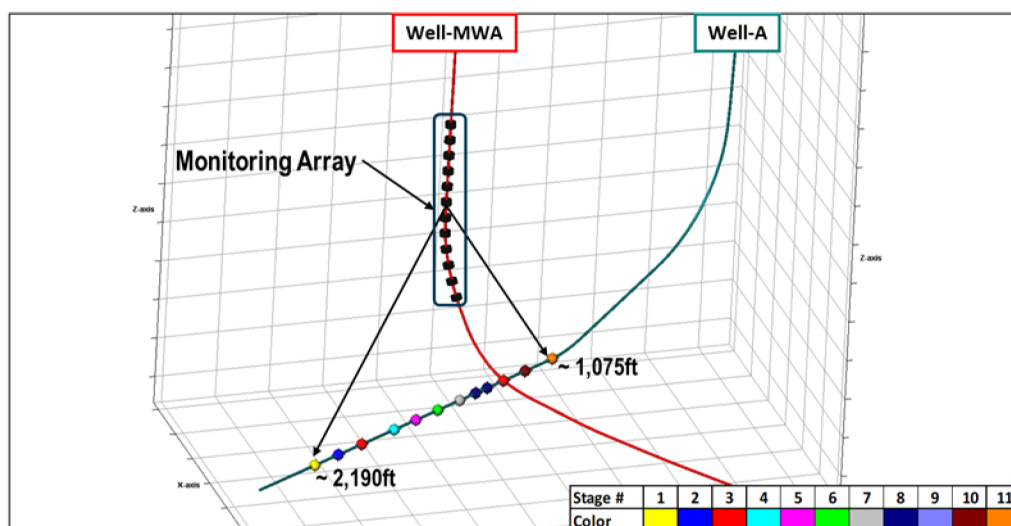


Figure 6—Monitoring setup for Well-A with distance from the middle of the monitoring array in Well-MWA to the perforations for first and last stages

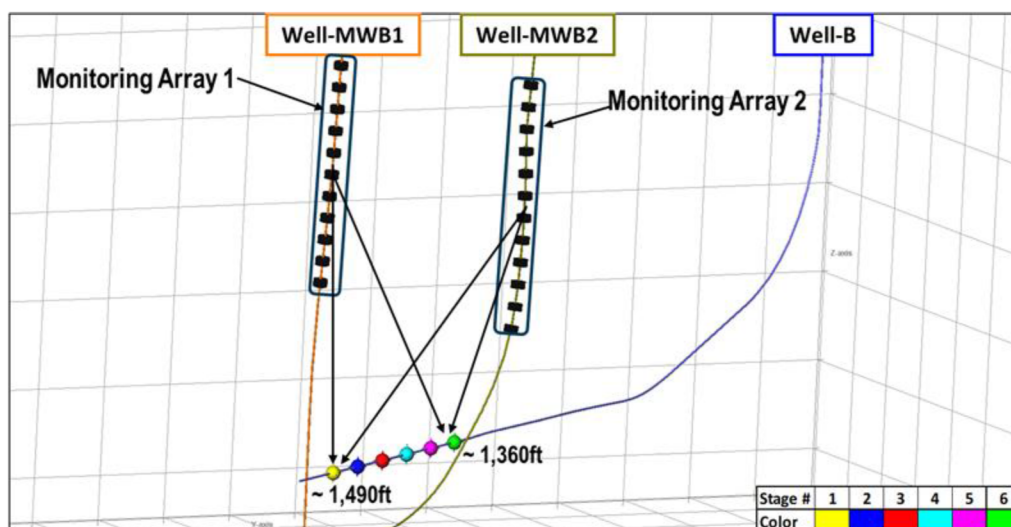


Figure 7—Monitoring setup for Well-B with distance from the middle of monitoring arrays in Well-MWB1 and Well-MWB2 to the perforations for Stage 1 and Stage 6

Tracer program

The development of hydrocarbon (oil or gas) chemical tracers applied in the form of an unstable emulsion with fracture fluids during each stage provides information on specific stage oil and gas production, providing operators additional insights to improve their completion and stimulation programs based upon actual hydrocarbon positional flow. The underlying principle of using hydrocarbon based chemical tracers in fracturing stages is to confirm stage fracture oil/gas production from each stage or section of the wellbore. The presence of the hydrocarbon tracers in produced oil or gas and their relative mass flowback versus time or hydrocarbon production volume provides an indication that all stages are open to flow in the wellbore with no obstruction from the milled plugs and which section or geology is most productive.

The technology uses hydrocarbon-based tracers in the form of unstable emulsions that are added to the aqueous stimulation fluid of each fracturing stage. The tracer is applied with the leading PAD viscous fluid and is transported deep into the fracture matrix via the fracturing gel. The selected specialized tracers are hydrophobic but have a very strong affinity for either hydrocarbon liquid or gas. Wherever the fracturing gel contacts hydrocarbon in the fractures, the tracer will move across from the aqueous phase into the in-situ

hydrocarbon. If hydrocarbon moves from each stage during well production, it will transport the specific tracer placed in the stage into the main wellbore and up to the surface. Samples of the produced hydrocarbons or gas are collected from the wellhead and analyzed for the presence and concentration of each tracer. A comparative analysis of the amount of tracer from each stage in the samples can then be related to the amount of hydrocarbon produced from that stage with more hydrocarbon produced from that stage resulting in more tracer removal from the stage (Spencer et al. 2013)

Tracer analysis was conducted to evaluate the stage-wise production contributions and fluid placement efficiency in Well-A and Well-B. Hydrocarbon-based tracers were injected into all the stages during the acid fracturing treatment, and their production was monitored over time.

Geomechanical modeling from offset cores

Rock mechanical testing was initiated following the stimulation of two horizontal wells and prior to a broader development campaign in the field. The initial stimulation results were inconclusive, prompting several key questions:

- What is the optimal landing zone for future horizontal wells?
- Where should fracture initiation be targeted for maximum effectiveness?
- What net pressure can be safely applied without risking fracture propagation into the underlying depleted water zone, which is separated from the main target interval by an approximately 70 ft barrier?

Ten depth points were selected within the target formation for mechanical testing. Each point was assigned a representative label based on its geological or structural context—such as landing zone 1 or 2, upper or lower barrier, tight limestone, mudstone, wackestone, and others. At each depth, the testing program included unconfined compressive strength (UCS), multistage compressive testing, Biot's constant determination, and Brazilian tensile testing. To evaluate rock anisotropy, both vertical and horizontal core plugs were prepared for selected depths. For triaxial testing, the confining pressure was defined as the mean effective stress specific to each depth point.

Unconfined compressive strength (UCS) tests ($P_c = 0$) revealed significant variability in properties between horizontal and vertical samples across zones 1 to 10. According to the laboratory, the core material was highly heterogeneous and fractured, making it difficult to obtain ideal paired samples. Several zones exhibited varying bulk densities, limiting their reliability for assessing anisotropy. Moreover, the observed strength differences may have resulted from the unconfined testing conditions, where rock behavior is strongly influenced by microstructural features, potentially leading to misleading interpretations.

Multistress compressive (MSC) testing at $P_c = 2900$ psi was performed on horizontal, vertical, and 45° samples from zone 1. Results indicated very low anisotropy: horizontal Young's modulus was comparable to the vertical, and Thomsen parameters (γ , ϵ) were near zero (Thomsen 1986, Horne 2013). Triaxial compression (TXC) tests at $P_c = 2900$ psi showed limited variability in mechanical properties, supporting the conclusion of isotropic elastic behavior under confining stress.

Based on the laboratory data, the formation exhibits low transversely isotropic vertical (TIV) anisotropy but is characterized by high heterogeneity.

Test results indicate that Biot's coefficient varies across the zones investigated, with values ranging from 0.7 to 1. No strong correlation was observed with rock properties; a weak negative correlation was noted with both bulk density and Young's modulus. A better correlation was found with log-derived porosity (Hall 2010). The observed variability in Biot's coefficient translates to an approximate 0.1 psi/ft difference in the estimated stress gradient.

Measured Young's modulus (YM) in the tested intervals ranges from 3 to 8 million psi (Mpsi), averaging approximately 5 Mpsi in landing zone 1 and 8 Mpsi in landing zone 2. In the current stress settings, zones

with elevated Young's modulus generally promote more effective fracture initiation and propagation during acid fracturing. These zones fracture at lower pressures and develop acid-etched channels that remain structurally stable upon closure. Conversely, softer or more heterogeneous zones tend to produce diffuse fracture or wormhole networks, which may fail to sustain fracture conductivity due to insufficient stiffness and cohesion. This observation is supported by production results from two stimulated wells. The well with the horizontal section landed in zone 1 (average YM ~5 Mpsi) showed lower production compared to the well landed in zone 2 (average YM ~8 Mpsi). Although microseismic data indicated that the entire formation was stimulated in both cases, the fracture initiation point appears to be a critical factor influencing stimulation effectiveness. Further investigation is required before drawing definitive conclusions.

Stress modeling, based on calibrated rock mechanical property profiles, was validated using closure pressures observed during diagnostic fracture injection tests. The barrier separating the main stimulation target from the underlying depleted zone exhibits elevated horizontal stress, attributed to its preservation of original pore pressure due to extremely low permeability Fig. 8. This stress contrast effectively limits downward fracture propagation. Microseismic measurements further confirm that fracture growth into the underlying interval was minimal.

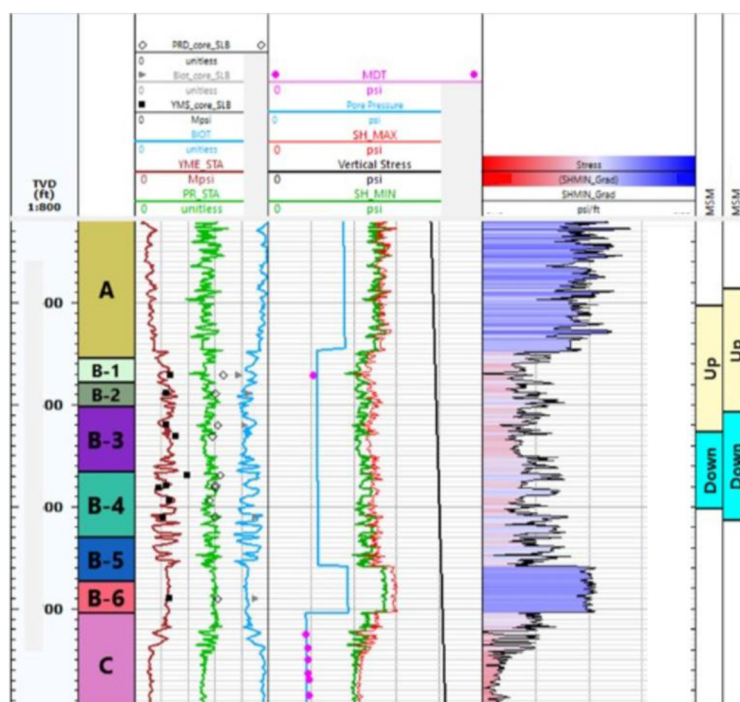


Figure 8—Mechanical properties and stresses for offset well, approximate fracture dimensions from wells A and B in the last two tracks

Treatment Execution

The treatment simulations resulted in ~4,000 psi surface pressure during execution. The XMAS tree pressure rating in this rigless operation is 3,000 psi. Accordingly, an XMAS tree isolation tool is required to protect and isolate the XMAS tree from the expected pressure exceeding the limitation. This results in continuous rig-up and rig-down of both wireline and fracturing wellhead equipment to complete the designed treatments. Simultaneously, the pumps injecting the tracers into the treatment fluids were rigged up to the fracturing surface treating lines. This demanded crucial and timely coordination between the different operations for the safe and successful delivery of the treatments.

The first Multistage Acid Fracturing treatment in Well-A was successfully executed placing 11 stages across the horizontal well using the PnP technique. A 28% Lightweight SPRA was utilized in 9 out of the

11 stages combined with 28% HCl, while the remaining 2 stages employed straight 28% HCl as per design for the purpose of evaluating the different treatment designs. Each stage was perforated with a single 5-foot cluster, apart from stage 8, which was specifically designed with double the treatment volumes to assess its impact on creating two hydraulic fractures with limited entry perforations and production from such design. Furthermore, low-temperature PDD were strategically deployed in Stage 8 to facilitate cluster diversion and support creating two hydraulic fractures. The success of the diversion was evident through observed pressure differentials and distinct pressure behavior. Fig. 9 details the volumes per stage in Well-A.

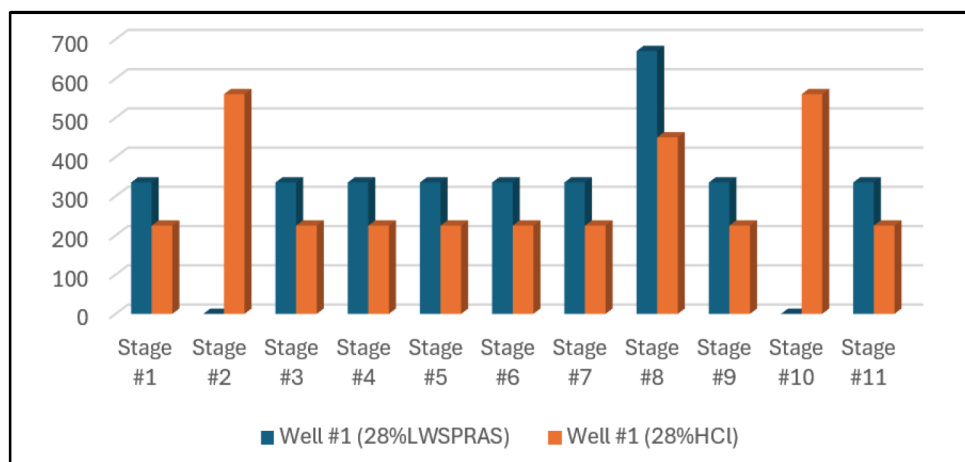


Figure 9—Well-A Treatment Fluids Volume by Stage

An analysis of the pressure decline plot for Well-A reveals that each stage exhibits a distinct Instantaneous Shut-In Pressure (ISIP), indicating variations across zones in each stage. Notably, Stage 2 demonstrates the steepest pressure decline, signifying the highest fluid leak-off, whereas Stage 8 shows the least decline, suggesting minimal leak-off. Meanwhile, the remaining stages display relatively consistent pressure decline behavior, reflecting similar characteristics across these stages, as observed in Fig. 10.

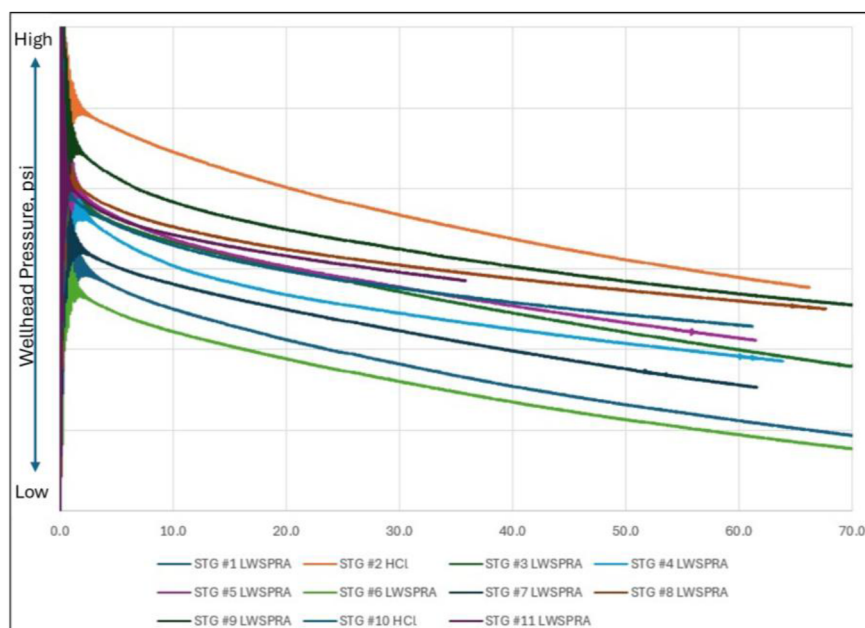


Figure 10—Well-A Treatment Pressure Decline

Additionally, three calibration injection tests were conducted in Stages 1, 5, and 9 before stimulation treatments. The derivative plots (Fig. 11, Fig. 12, and Fig. 13) of these 3 tests at shut-in were used to evaluate the fracture closure pressure, viscous PAD fluid efficiency, and near wellbore effects. These results were utilized to further calibrate the geomechanical model and re-simulate the acid fracturing treatment design. Derivative analyses of Stages 1 and 9 indicate height recession and a clear estimated of fracture closure. However, derivative analysis of Stage 5 shows Pressure Dependent leak-off response and a slightly lower closure pressure estimate which is within range of the other 2 tests in Well-A.

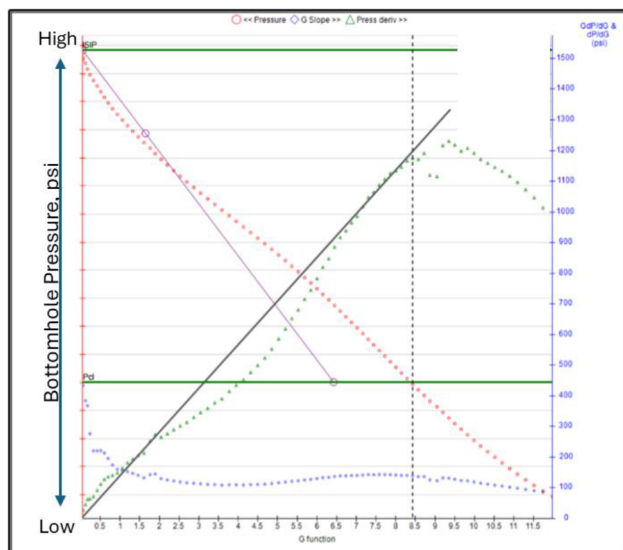


Figure 11—Calibration Injection Test Derivative Plot of Stage 1

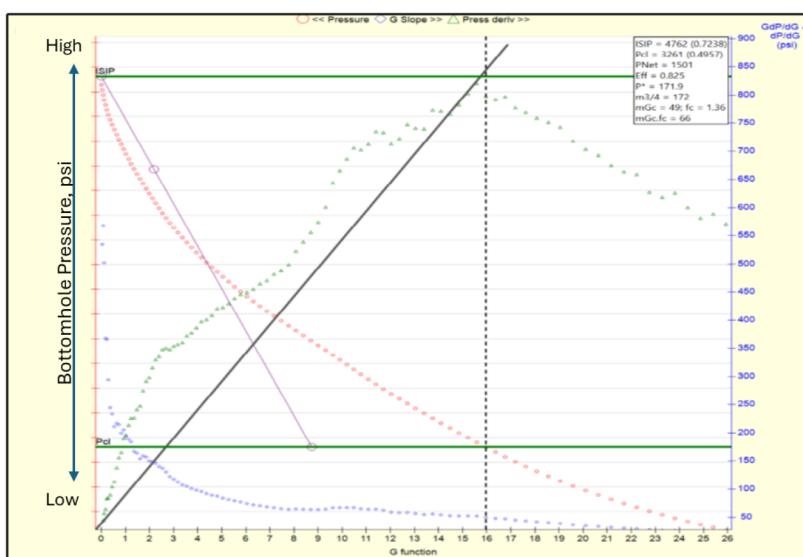


Figure 12—Calibration Injection Test Derivative Plot of Stage 5

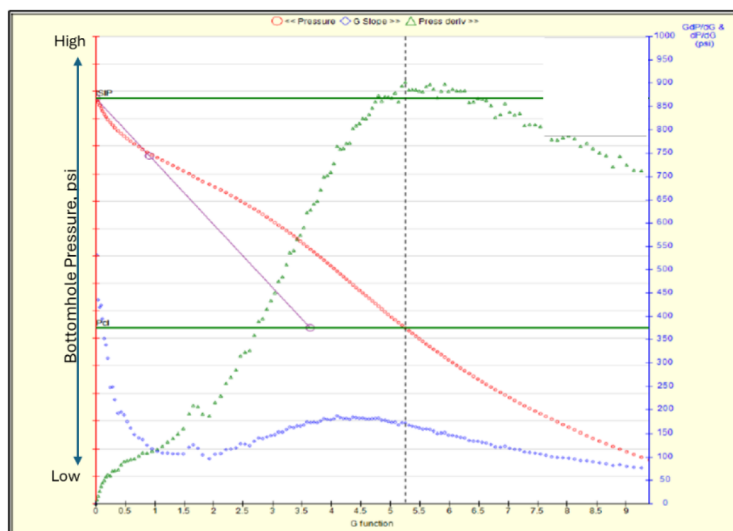


Figure 13—Calibration Injection Test Derivative Plot of Stage 9

The second Multistage Acid Fracturing treatment in Well-B consisted of 12 stages across the horizontal well, utilizing the PnP technique. The 28% Lightweight SPRA combined with 28% HCl were employed in 6 out of 12 stages, while the remaining 6 stages used straight 28% HCl. The designs were alternately pumped, starting with straight 28% HCl in Stage 1, followed by Lightweight SPRA in Stage 2, and continuing this pattern throughout the 12 stages. This alternating design was implemented to compare the fracture geometry and production performance between the two treatment designs while maintaining the same total acid volume per stage, as shown in Fig. 14.

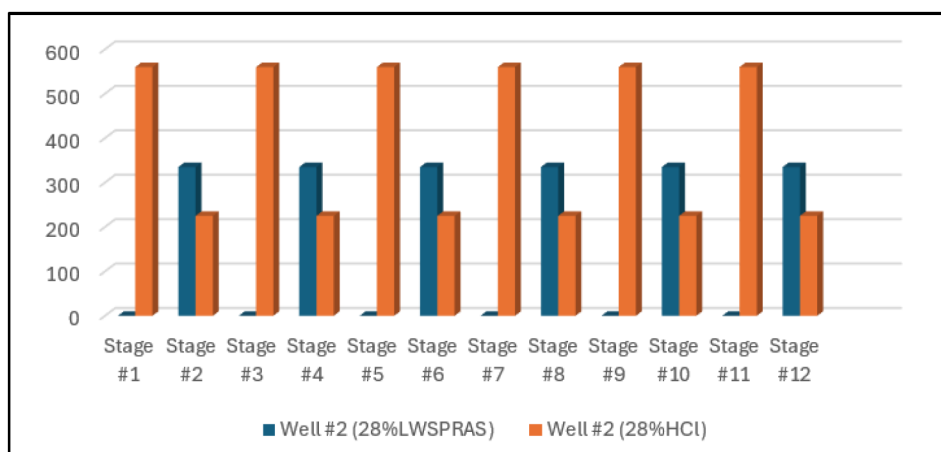


Figure 14—Well-B Treatment Fluids Volume by Stage

The pressure decline plot for Well-B indicates that each stage displays unique Instantaneous Shut-In Pressure (ISIP) values, reflecting variations across zones in each stage. Stages 11 and 12 exhibit the steepest pressure drops, indicating significant fluid leak-off, whereas Stage 3 shows the gentlest decline, suggesting minimal leak-off. On the other hand, the remaining stages demonstrate comparable pressure decline trends, indicating similar behavior across these stages, as per Fig. 15.

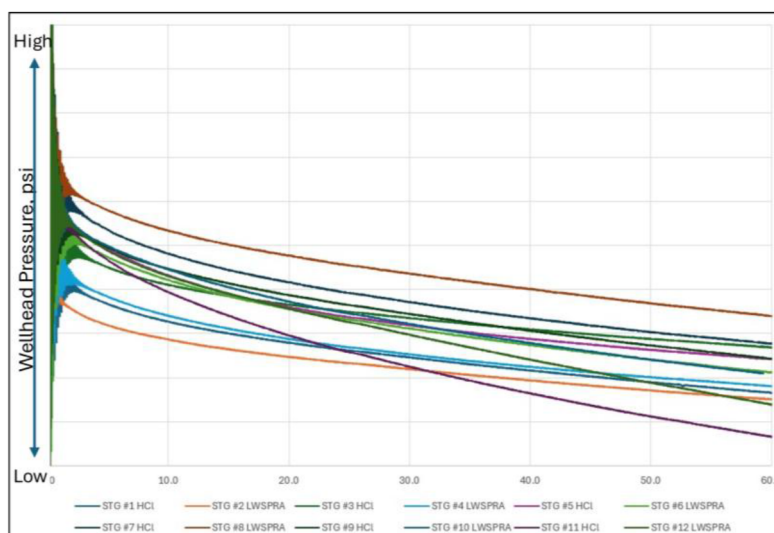


Figure 15—Well-B Treatment Pressure Decline

Evaluation & Results

Fracture Geometry Comparison from Microseismic Data

Well-A

The microseismic data from the first three stages on Well-A was associated with a very low event count due to the longer monitoring distance from the array. The events over the course of pumping were not sufficient to extract the fracture dimensions for these stages. As a result, dimensions were extracted only for eight stages (Stages 4 to 11). Of these eight stages, only one stage (Stage 10) was executed with the straight 28% HCl.

Fig. 17 shows the microseismic events colored by stage along with their dimensions in map view. The fracture orientation/azimuth was observed to be in the range of N35-40°E for the majority of the stages. In Stage 10, which utilized straight 28% HCl, the fracture azimuth exhibited a N25°E trend. The average microseismic length for these stages was 872ft with Stage 10 having a significantly longer length of 1456ft (Fig. 18). The average fracture height from the microseismic data was 227ft with an upwards height of 121ft and downwards height of 106ft (Fig. 19). Stage 10 was associated with a localization of activity in one fracture bi-wing which impacted the azimuth. In microseismic data, localization of activity with significant event count in one particular region is usually associated with a subsurface feature/structure which influences the fracture orientation and propagation. In Well-A, Stage 10 had an event count close to two times the stage with the next highest event count (Fig. 20). These observations suggest that Stage 10 could have been impacted by a subsurface feature that contributed to an associated higher event count and larger microseismic length when compared to other stages.

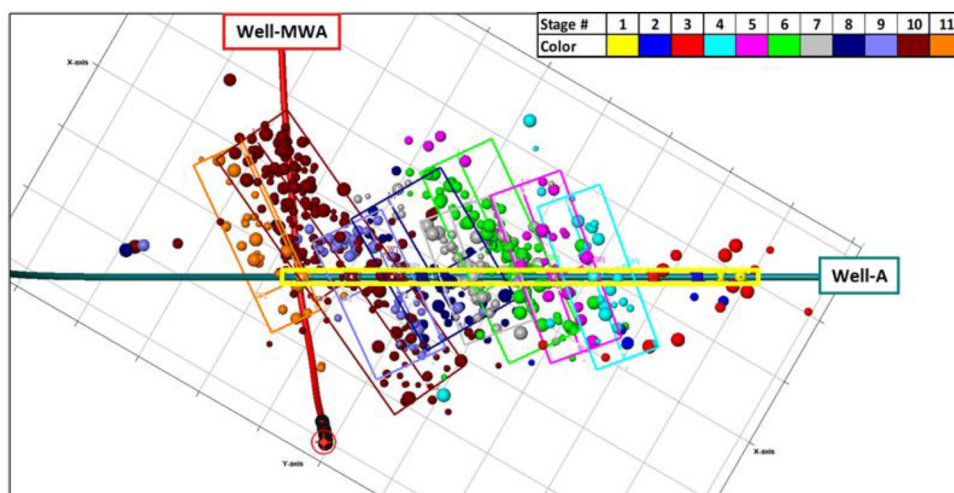


Figure 16—Map view of fracture dimensions for Stages 4 to 11 in Well-A with events and dimensions color coded as per the legend in the image

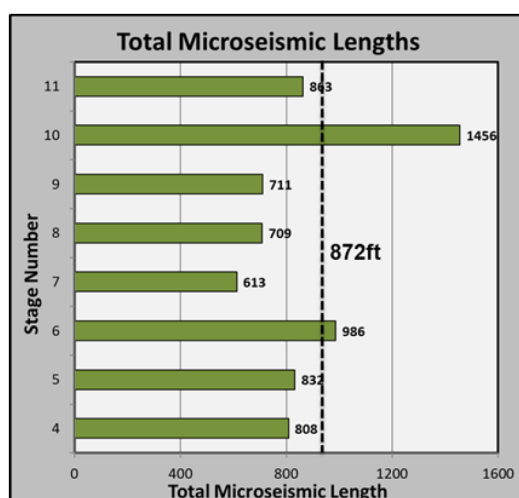


Figure 17—Summary of fracture length extracted from microseismic data by stage for Stages 4 to 11

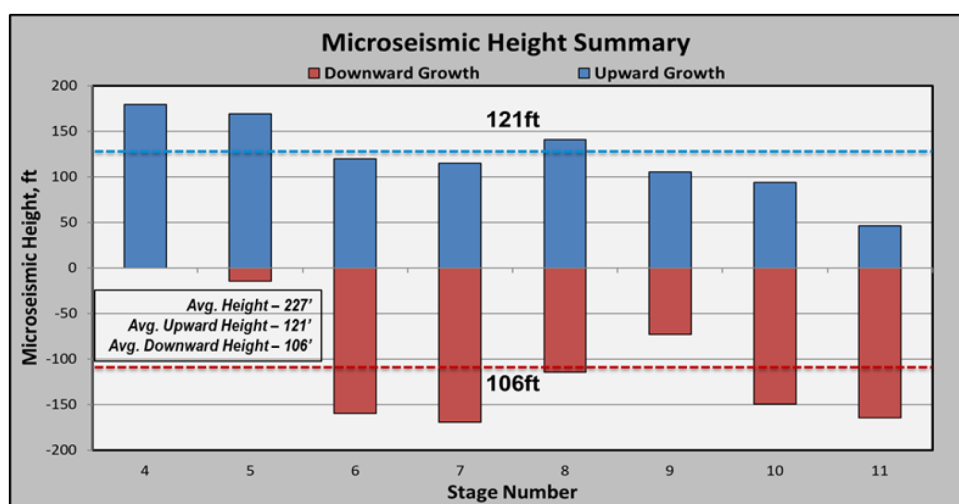


Figure 18—Summary of fracture height extracted from microseismic data by stage for Stages 4 to 11 with upwards and downwards height from the perforations

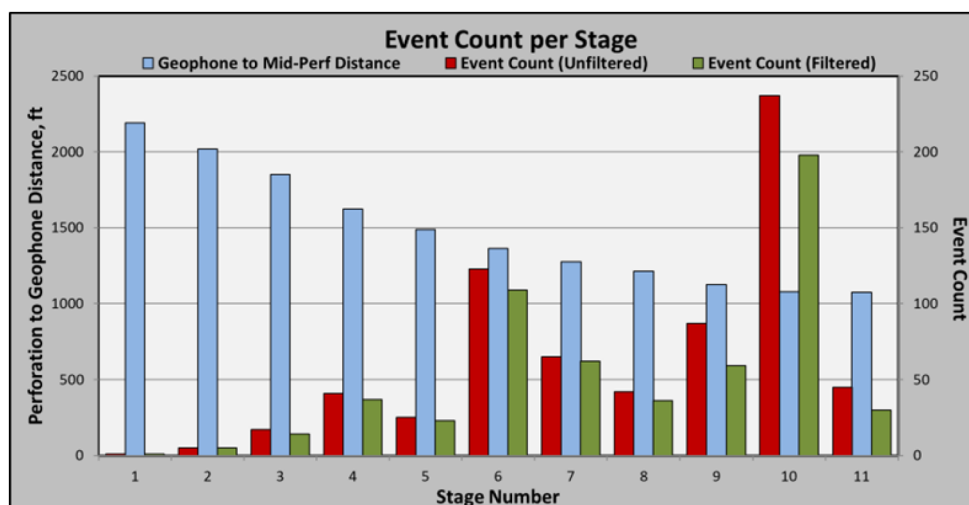


Figure 19—Event count by stage for the monitored stages along with the monitoring distances from the array to the perforations. Stage 10 had a significantly higher event count compared to other stages

Since only one stage that utilized 28% HCl had sufficient events for fracture dimension extraction and combined with the trend observed with respect to localization of events, a comparison of the two acid systems was not performed for Well-A.

Low Temperature PDD Evaluation with Microseismic Data. Stage 8 diversion effectiveness was also evaluated using microseismic data in addition to the pressure response. During the first half of the stage prior to the Low Temperature PDD, the microseismic events exhibited fracture orientation and propagation trends like what was observed in the previous stages. After the Low Temperature PDD, from the microseismic data, no significant activity was observed in the first cloud from the treatment prior to the diversion. The lack of overlap of events from second half of the treatment post Low Temperature PDD in the first cloud could imply that the fracture growth was in a different region due to diversion.

Well-B

Microseismic monitoring was done for the first 6 stages in Well-B which was completed using straight 28% HCl and 28% Lightweight SPRA with 28% HCl in an odd-even sequence. Stages 1, 3, and 5 employed straight 28% HCl while Stages 2, 4, and 6 employed 28% Lightweight SPRA with 28% HCl. There was sufficient event count in all the monitored stages for hydraulic fracture dimension extraction.

Fig. 20 shows the microseismic events colored by stage along with their dimensions in map view for the first six stages. The fracture orientation/azimuth was noted to be N45°E for all the monitored stages. The average microseismic length for the stages was 809ft (Fig. 21) while the average height from the microseismic data was 199ft (Fig. 22). The upwards height was 124ft while the downwards height was 75ft for the 6 stages.

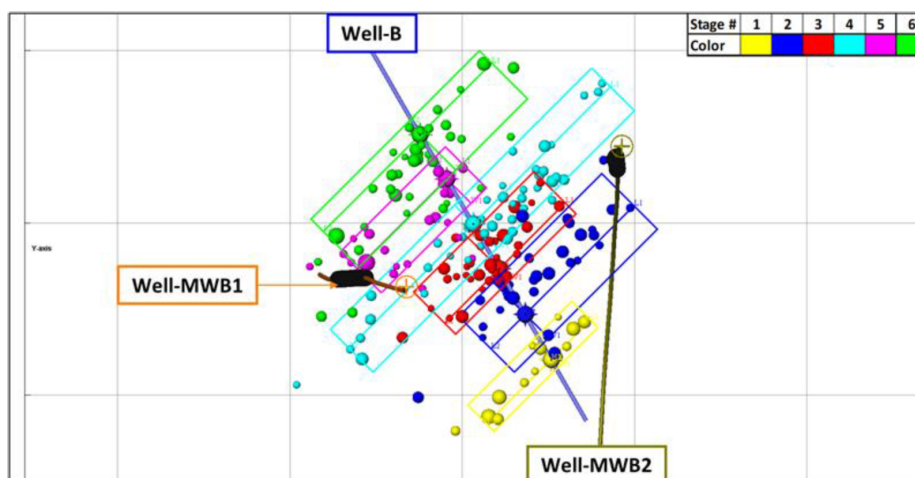


Figure 20—Map view of fracture dimensions for Stages 1 to 6 in Well-B with events and dimensions color coded as per the legend shown in the image

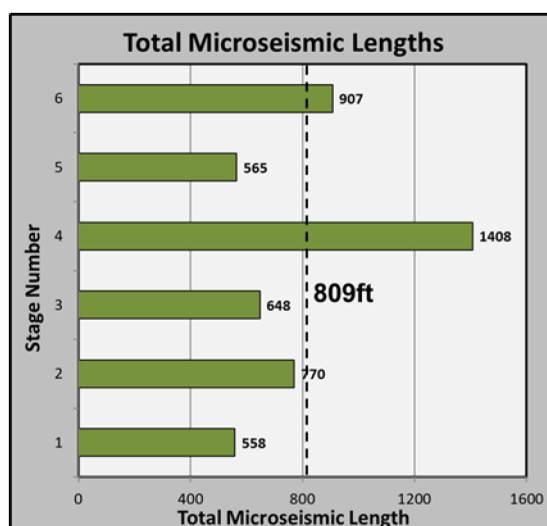


Figure 21—Summary of fracture lengths extracted from microseismic data by stage for Stages 1 to 6

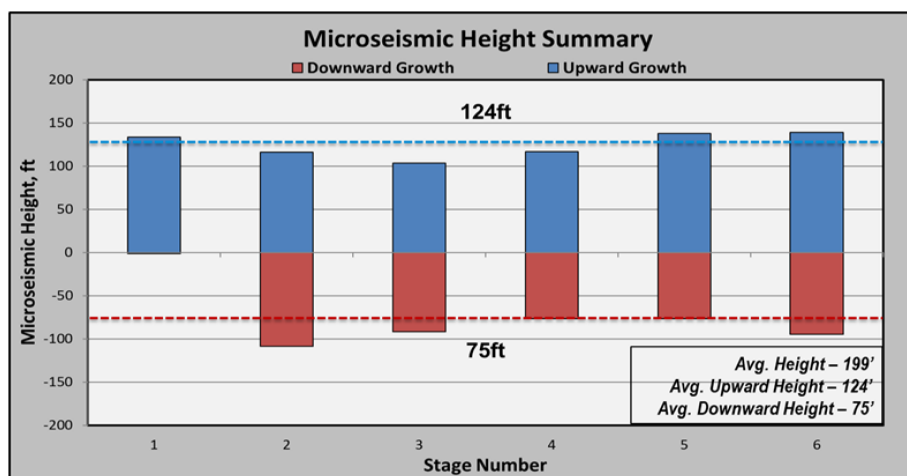


Figure 22—Summary of fracture height extracted from microseismic data by stage for Stages 1 to 6 with upwards and downwards height from the perforations

While comparing the straight 28% HCl and 28% Lightweight SPRA stages, there was no notable difference in the fracture orientation/azimuths. All the monitored stages were observed to have an event cloud orientation of N45°E. For vertical extension of the fractures, both fluid systems exhibited similar fracture heights in the upward direction. However, the Lightweight SPRA stages had deeper events below the perforations when compared with the straight 28% HCl stages. In terms of microseismic length, a significantly higher length extension was observed for the Lightweight SPRA stages. The microseismic dimensions summary by fluid system is summarized in Table 1. Fig. 23 shows the microseismic events for stages colored by fluid system in map view and depth view where the larger length extension and slightly deeper height extension for Lightweight SPRA stages can be visualized.

Table 1—Microseismic Dimensions summary for the 28% HCl and 28% Lightweight SPRA stages

Fluid System (Stages)	Average Microseismic Length (ft)	Average Microseismic Height (ft)	Azimuth (°)
28% HCl (Stage 1,3,5)	590	182	N45T
28% LWSPRAS (Stage 2,4,6)	1028	217	N45°E

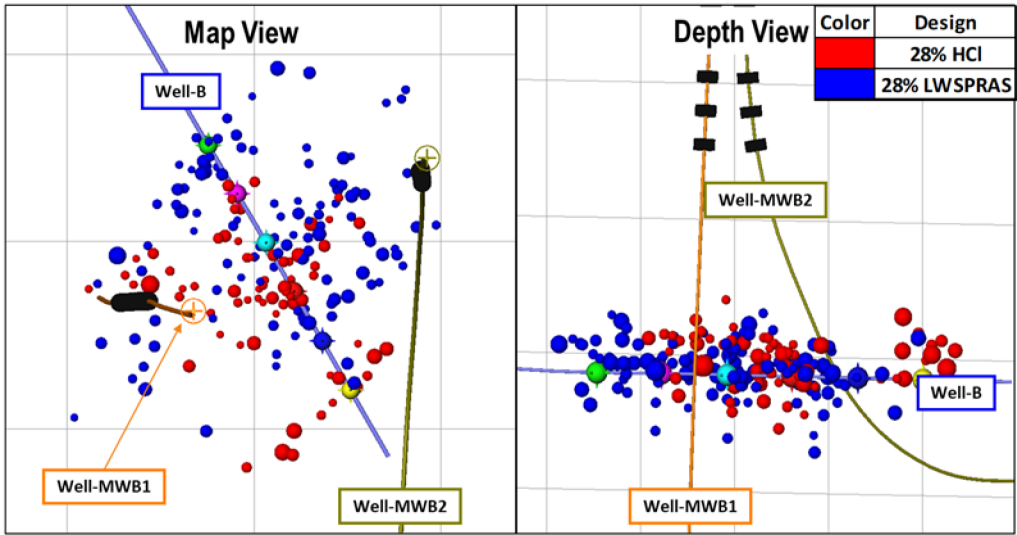


Figure 23—Microseismic events colored by fluid system for the monitored stages in Well-B in map view (left) and depth view (right)

Fracture orientation complexity

Analysis of the orientation of the acquired microseismic events reveals complex interactions between the induced fracture during the stimulation and the structural features identified by the individual sonic imaging reflections in 3 dimensions. Fig. 24 illustrates an overlay of microseismic events captured during Stage-9 stimulation and extrapolated sonic imaging structural features in Well-A. The microseismic events are represented by dark blue-colored spheres. The 3D sonic imaging high angle events are extrapolated and plotted as colored lines (sticks) based on their compliance. Early microseismic events are seen to propagate in the regional SHMax direction towards the left of the wellbore. The asymmetric propagation of the induced fracture is further accentuated by the strongly compliant structural feature near the perforation. Late time microseismic events appear to be due to re-activation of the non-compliant features on the right of the wellbore during the stimulation treatment.

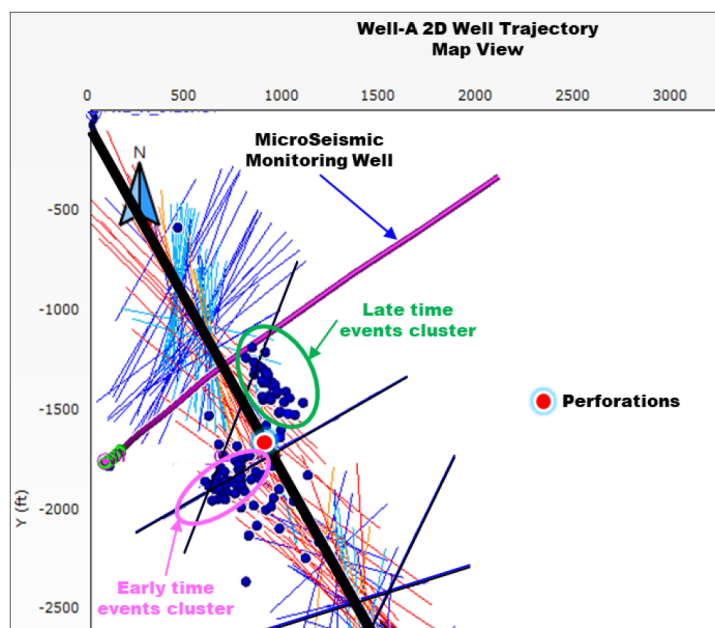


Figure 24—Well-A Microseismic Events and 3D Sonic Imaging structural features Overlay from Stage-9

Fig. 25 shows an overlay of all the microseismic events with the 3D sonic imaging structural features in Well-A and Well-B. A few conclusions based on the observations can be drawn as below:

- Majority of the microseismic events are aligned with the SH_{Max} direction. These are believed to be due to the induced fracture propagation and some leak-off into the pre-existing compliant structural features.
- Microseismic events are also seen parallel to partially compliant structural features. These are believed to be due to re-activation of the features during the dynamic pore pressure variations during the stimulation treatment.
- Strongly non-compliant features appear to be restricting induced fracture propagation under in the injection pressure limits.
- Stages with a high density of non-compliant fractures show "Height Recession / HF Misalignment" signature in pre-frac diagnostic injection treatments.
- Stages with a high density of compliant fractures show "Pressure Dependent Leakoff" signature in pre-frac diagnostic injection treatments.

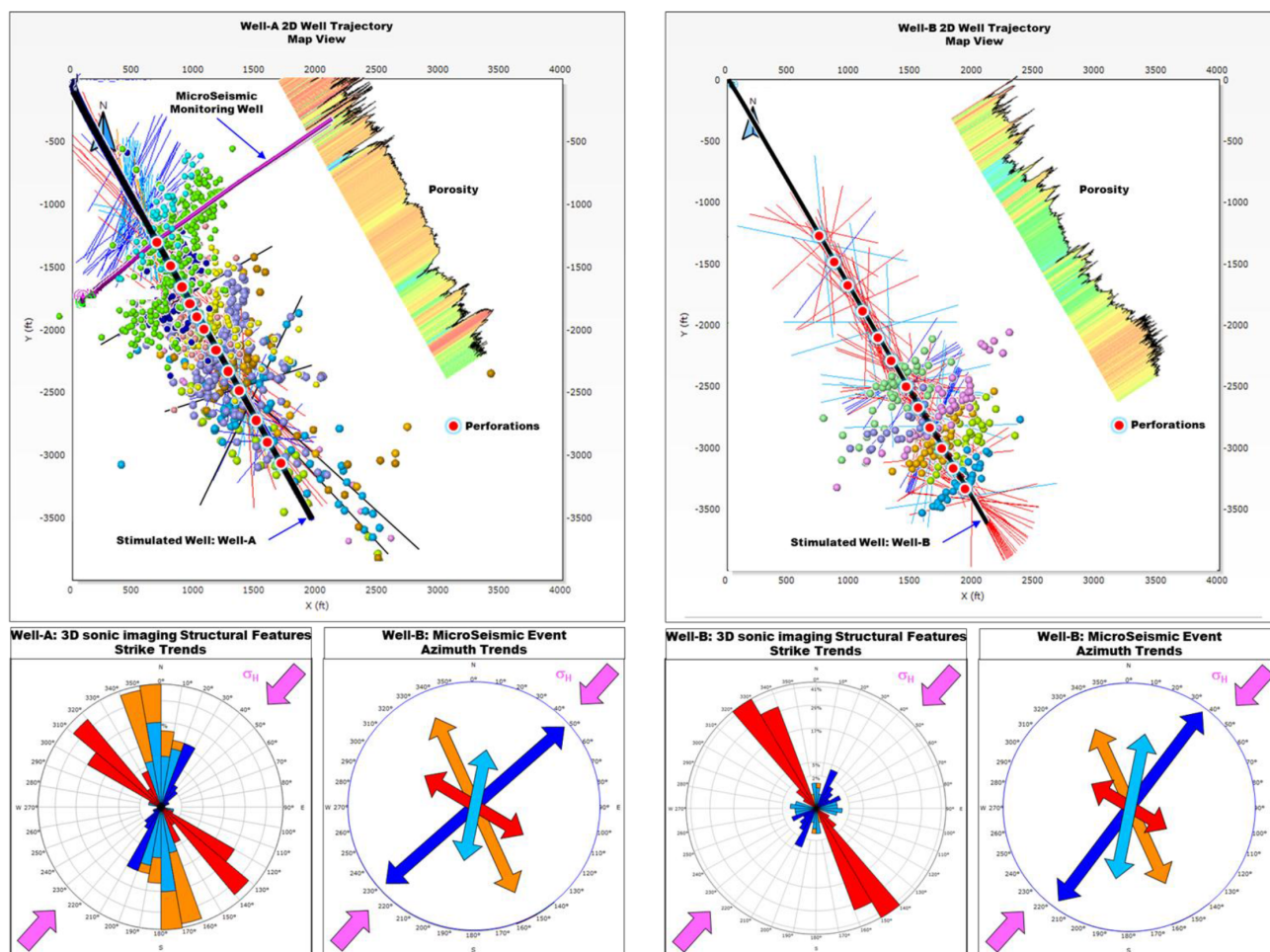


Figure 25—Microseismic Events Overlay with Far Field 3D Sonic Imaging structural features in Well-A & Well-B

Tracer distribution

The results provided valuable insights into the effectiveness of the acid systems and the stimulated reservoir volume. Stage-wise production data revealed variations in oil production rates and cumulative oil volumes, highlighting differences in the performance of the acid systems used. The tracer data also indicated broader stimulation coverage and improved fluid efficiency in stages treated with the Lightweight SPRA compared to 28% HCl. By quantifying the production contributions from individual stages, the tracer results played a critical role in assessing the overall success of the acid fracturing campaign and guiding future stimulation designs.

The plots of oil tracer production graphed over 9 weeks in Well-A and 9 weeks in Well-B show very different production profiles from each wellbore as seen in Fig. 27 and Fig. 27. A total of 53 and 48 samples were collected from Well-A and Well-B, respectively. They are particularly useful in understanding how oil production changes over time from each stage.

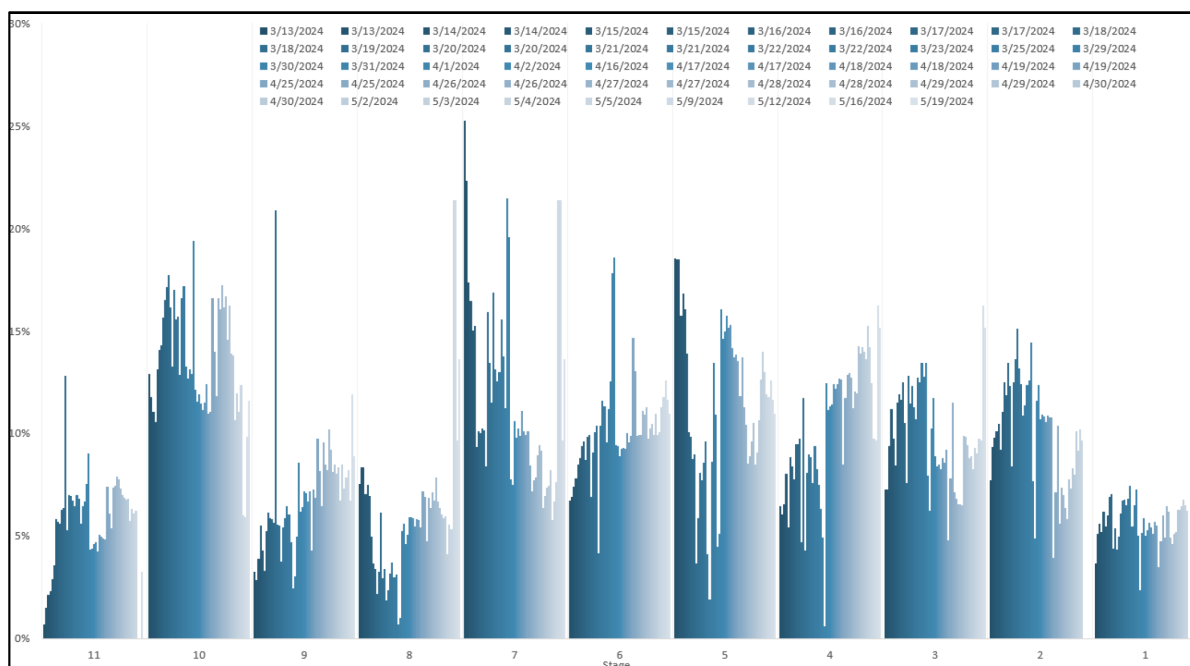


Figure 26—Well-A Stage Contribution Over Time

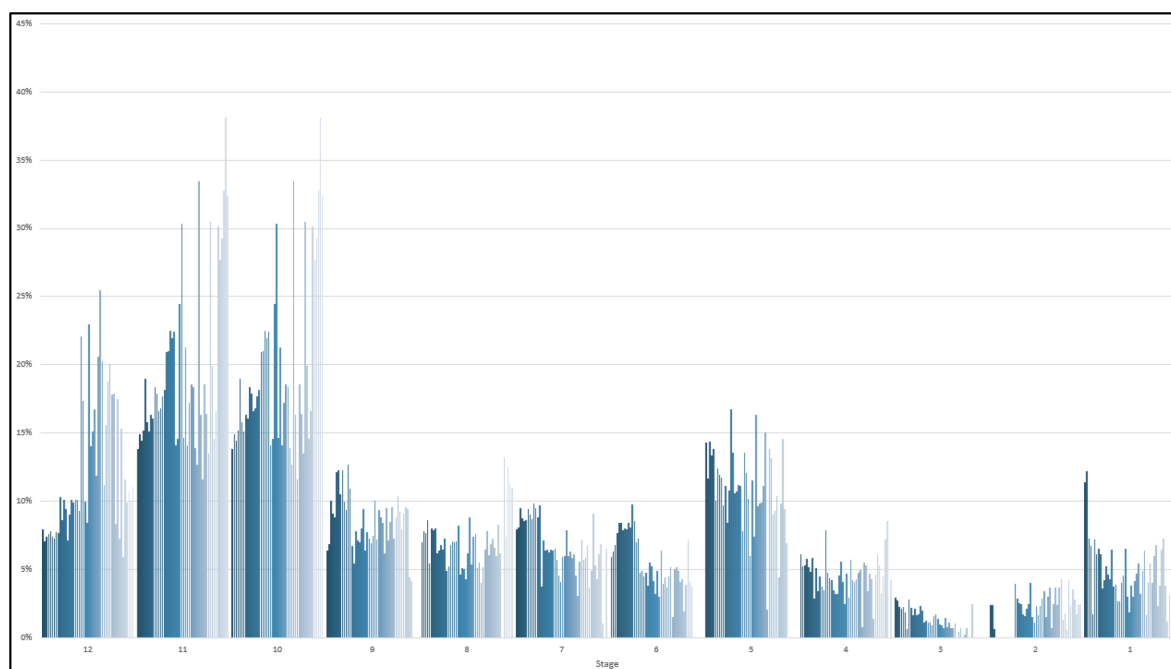


Figure 27—Well-B Stage Contribution Over 9 Weeks Period

Microseismic Findings

There was only one stage that employed the straight 28% HCl fluid system where dimensions were extracted in Well-A. This stage was also associated with a subsurface trend from microseismic data which was not observed in the other stages. Hence, a comparison of fluid systems could not be performed for this well. In Well-B, where there was an alternating sequence of both fluid systems, comparison was made for the 6 monitored stages. There was no significant difference observed in the fracture orientation for the fluid systems. Similarly, only a slight variation in the downwards height growth was observed in the 28%

Lightweight SPRA system. In terms of microseismic length, the extension of the 28% Lightweight SPRA stages was significantly larger across the monitored stages.

Conclusion and Way Forward

This study plays a crucial role in selecting the landing point of future drilling across this thick tight carbonate reservoir. The findings from far field 3D sonic imaging combined with microseismic analysis uncovered the existence of natural fractures in the reservoir. Based on the estimated long fracture half-length from microseismic events, it concludes that 28% acid strength over achieves the necessary fracture half-length and hence, it is possibly too strong for such a heavily fractured low temperature reservoir. Together with the rock analysis and production results, it is possible that there's less connectivity at the near wellbore region.

Therefore, as a way forward, refine the future stimulation design by utilizing the updated geomechanical model, calibrated by the two completed wells, as well as changing the landing point to zone 3. A combination of gelled acid and Lightweight SPRA with acid strength in the range 20% is a possible alternative to the acid fracturing treatment design, maintaining the same fluid volumes, to achieve less rock dissolution at near wellbore and shorter, more optimal etched frac half-length. Future evaluation would include building a Discrete Fracture Network, and a pressure match from the stimulation and geomechanics.

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